1. Solve the following equations by factorisation

(a) $x^2 + 2x - 35 = 0$ (b) $x^2 - 15x - 54 = 0$

(c) $x^2 - x - 90 = 0$ (d) $x^2 + 15x + 54 = 0$

(e) $x^2 + 20x + 51 = 0$ (f) $x^2 - 12x + 32 = 0$

(g) $x^2 - 24x + 143 = 0$ (h) $x^2 - 17x + 60 = 0$

(i) $x^2 - 14x - 176 = 0$ (j) $x^2 - 26x + 133 = 0$

(k) $x^2 + 7x - 44 = 0$ (l) $x^2 + 2x - 195 = 0$

(m) $2x^2 - 5x + 3 = 0$ (n) $2x^2 - 7x - 9 = 0$

(o) $2x^2 + 13x + 6 = 0$

2. Solve the following equations

(a) $y^2 = y + 56$ (b) $12w^2 = 13w - 3$

(c) $11y = -4 - 6y^2$ (d) $c(c - 1) = 2$

(e) $q^2 = -2(q - 4)$ (f) $d(d + 2) = 3$

(g) $x(x - 5) = 84$ (h) $y(5y + 27) = 18$

(i) $3p^2 = 6p(2 + p)$ (j) $2x(4x + 5) = 3$

(k) $13x = 2(2x^2 + 5)$ (l) $2(10 - x^2) = 3x$

(m) $4y - 3 = 3y(y - 2)$ (n) $-12y - 9(y + 1) = 6y^2$

(o) $(a + 4)(a - 2) = -5$ (p) $(3x - 4)(x - 4) = -5$

3. Solve the following equations

(a) $x^2 = 25$ (b) $a^2 = 36$ (c) $y^2 = \frac{49}{4}$

(d) $b^2 - 16 = 0$ (e) $a^2 - 64 = 0$ (f) $x^2 - \frac{4}{81} = 0$

(g) $4y^2 = 9$ (h) $2x^2 = 32$ (i) $3p^2 - 27 = 0$

(j) $5p^2 - 20 = 0$ (k) $25b^2 - 40 = 9$ (l) $3b^2 - 8 = 4$

4. The area of a parallelogram is $50\ cm^2$. If the base is twice its height, calculate the height.

5. The breadth of a rectangular plot of land is $5\ m$ less than its length. If the area of the plot is $104\ m^2$, find the dimensions of the plot.

Advanced Questions

6. Solve the following equations

a) $(3x+2)(2x+1)=1$
b) $9n^2-27n+20=0$
c) $(3x-4)(x-2)-5=0$
d) $2(4x+1)(x-1)+3=0$
e) $(3x-2)(3x+1)=-\frac{1}{2}(21x+6)$
f)
$$x-2=\frac{4}{x+1}$$
g)
$$3x-1=\frac{4}{2x+1}$$
h)
$$\frac{4}{9}-\frac{d^2}{25}=0$$
i) $(3s+1)(s+4)=-5(s-1)$
j)
$$\frac{1}{2}x^2-\frac{11}{4}x+\frac{5}{4}=0$$
k)
$$\frac{3}{x+1}-\frac{1}{2}=\frac{2}{3x-1}$$
l) $\sqrt{2}x^2+7x+5\sqrt{2}=0$
m) $3a^2x^2+8abx+4b^2=0$, where $a \text{ and } b \text{ are constants and } a\neq 0$
n)
$$\frac{3x}{x-1}+\frac{1}{x-2}+\frac{4x-1}{(x-1)(x-2)}=0$$